Product Brief: RAG-Diary

> A short business requirements document, written before the Software Requirements Specification. It sets out the problem, the goals and the boundary of the first release.

> Thai version: PRODUCT-BRIEF.md

1. Document control

Item	Detail
Document	Product Brief (business requirements) for RAG-Diary
Version	1.0
Prepared	15 October 2026
Author	Product Analyst (project owner)
Approver	Product Owner
Status	Reviewed, round 1 (25 October 2026)
Related documents	SRS, Risk Register, Roadmap, DFD-Privacy

2. Executive summary

RAG-Diary is a web application for a private diary that addresses one stubborn problem of long-term journaling: people write, and later cannot find what they wrote. Most people write in their own words, but when they want to look back they cannot recall the words they used, so keyword search performs badly.

The product uses retrieval-augmented generation so that people can retrieve old entries by meaning rather than by exact wording. From there it extends into a conversation with one's own notes, and into a summary of the mood of a period, expressed as text and as an image, so a memory worth keeping takes a form you can hold.

The target users range from occasional writers to people who journal as a routine and want a tool for long-term self-review. Privacy comes first: each person's data is isolated with row-level security, and no user can see another user's entries.

3. Business problem

ID	Problem	Effect on the user
P-01	Keyword search on old entries returns results that miss the intent	The memory cannot be found, and the effort of writing loses value
P-02	Entries pile up with no way to see the whole	Trends are invisible, such as a difficult stretch or a topic that keeps returning
P-03	The data is sensitive, so security and use are a worry	People avoid writing the important things for fear of exposure or misuse
P-04	Self-review takes too much effort	People stop writing over time because the return is not felt
P-05	Memories worth keeping have no form to be kept in	Important periods fade, because only raw text remains under newer entries

4. Goals and success metrics

Goal	Metric	Target
G-01 Users can retrieve memories by meaning	First-attempt searches that return the intended entry	At least 80 percent in user testing
G-02 Writing feels worth the effort	Share of users who write again within 7 days	At least 40 percent
G-03 Private data is isolated and controllable	Cases where one user sees another user's data	0 cases
G-04 The system responds fast enough to keep attention	Response time for semantic search	No more than 3.5 seconds at the 95th percentile
G-05 Users can keep a memory in a tangible form	Users who produce a mood summary and an image on the first try	At least 80 percent in user testing

> G-01, G-02 and G-05 are measured after user acceptance testing, which is the next step of the project.

5. Scope

5.1 In scope (version 1.0)

ID	Included	Business reason
IS-01	User accounts (sign up, log in, log out)	Required to isolate data per person

ID	Included	Business reason
IS-02	Create, edit and delete diary entries with a tag	The core function of the product
IS-03	Automatic semantic indexing of entries (embeddings)	The precondition for search by meaning
IS-04	Semantic search	Answers P-01 directly
IS-05	Question-and-answer chat with one's own entries (RAG chat)	Creates value for self-review
IS-06	Mood summary of a period as text and as an image	Answers P-02, P-04 and P-05
IS-07	Self-service export and deletion of all data	A privacy requirement (P-03)

5.2 Out of scope (version 1.0)

ID	Not included	Reason
OS-01	Sharing entries with others or with a community	Contradicts the privacy stance of the first release
OS-02	An admin role that can read user data	No real use case, and it breaks least privilege
OS-03	Reminders to write	A future scope extension, not the core problem
OS-04	A native mobile app	A responsive web app meets the first goal
OS-05	A third UI language beyond Thai and English	Version 1.0 ships Thai and English (CR-09), which covers the first audience

6. Users and personas

Persona	Role	Main need	Obstacle
Guest	Has no account, wants to see what the product does	Understand the value quickly and try it without commitment	Distrust of giving personal data to a new system
Casual user	Writes occasionally, enjoys the creative side	Write and generate an image easily, without a long session	Uses it rarely, so forgets how, and hesitates to share data
Reflective user	Journals as a routine, wants long-term mood review	Easy writing, reliable search, safe data	No time, and fear of a leak

Persona	Role	Main need	Obstacle
Admin or Ops	Keeps the system healthy	Visibility of system status	Not in version 1.0 (see OS-02)

7. Stakeholders

Stakeholder	Main interest
Product Owner	Value delivered to users and product direction
End user	Privacy, simplicity and speed
Development team	Requirements that are clear and testable
External AI provider	Use within the provider's terms, and cost per call

8. Assumptions and constraints

8.1 Assumptions

- A-01 Users accept that entry text is sent to an external AI provider in exchange for search and summaries. That consent must be explicit.
- A-02 Users reach the product through a current browser on a computer or a phone.

8.2 Constraints

- C-01 AI calls cost money per request, so call volume must be limited and misuse prevented. The limiter is implemented in the v1.1 code (the rate_limits table) and awaits the migration before it is active (see ROADMAP and RISK-REGISTER R-02).
- C-02 Some processing depends on external services (Supabase and the AI provider), whose timing cannot be fully controlled.
- C-03 Diary content is sensitive personal data and must comply with Thailand's Personal Data Protection Act (PDPA).

9. Business risks (summary)

ID	Risk	Level	Mitigation in brief
BR-01	Diary data leaks outside the user's boundary	High	Row-level security, with every data access point reviewed
BR-02	AI cost grows through repeated calls	Medium	Rate limiting and usage monitoring
BR-03	Users do not return because search misses their intent	Medium	Test with real users and tune the similarity threshold before launch

Full system risks and mitigation plans are in the Risk Register.

10. Definition of done (product level)

- Every in-scope function (IS-01 to IS-07) works and passes the acceptance criteria in the SRS.
- No case exists where one user can reach another user's data.
- Users can export and delete their own data without help.
- Text that leaves the system for external processing has been disclosed and accepted by the user.

11. References

Document	Relationship
SRS	Turns the business requirements here into testable specifications
TRACEABILITY-MATRIX	Shows where each requirement is designed and tested
RISK-REGISTER	Full risk detail and mitigation
DFD-PRIVACY	The flow of personal data and the PDPA analysis